# The Sky Has a Pulse

A living digital ecosystem of white stork movement, survival, and disappearance

DSC 106 Final Project Proposal

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## Project Title

**The Sky Has a Pulse**

## Dataset

We will visualize the public Movebank dataset *LifeTrack White Stork SW Germany (2013–2019)*, DOI: 10.5441/001/1.ck04mn78. The local dataset contains a reference table with 88 tagged white stork deployments and large accelerometer files with more than 6.3 million sensor events. The data is not synthetic, has far more than 100 rows, and includes timestamps, individual bird identifiers, acceleration readings, deployment metadata, sex, dates, and recorded deployment outcomes. For this proposal, we use the full reference metadata plus a systematic every-500th-row acceleration sample so the static prototypes can be generated quickly while preserving the temporal range of the study.

## Brief Writeup

This project will transform white stork accelerometer data into an explorable digital ecosystem. Instead of presenting a dashboard, the page will open as a dark living sky where each bird appears as a glowing pulse. Scrolling will move the reader from the full flock to seasonal currents, individual pulse orbits, and vanishing trails. Hovering near a pulse will reveal identity and motion; clicking a bird will fold the ecosystem into that individual stork's rhythm. The central question is: *what can we learn when migration is shown not only as route, but as bodily motion over time?* The final D3/Canvas experience will use animation, proximity-based hover states, a month scrubber, and narrative annotations to reveal patterns progressively. The intended takeaway is that animal tracking data is not just location or activity; it is a fragile archive of living rhythms, interruptions, and survival.

## Planned Interactive Experience

- **Opening hook:** a black sky slowly fills with glowing stork pulses; each pulse is an individual animal.
- **Scroll-driven narrative:** the reader moves from ecosystem scale to season scale to individual scale.
- **Tactile interaction:** pulses bend toward the cursor, hidden trails appear on hover, and clicking locks onto one bird.
- **Time exploration:** a month scrubber moves a beam of attention through the study period.
- **Emotional encoding:** cyan/amber light shows motion intensity; red fractures mark recorded death or disappearance.
- **Build plan:** D3.js for data joins/scales, Canvas or WebGL for particles, GSAP for scroll transitions, and optional Web Audio API for subtle wetland ambience.

## Static Visualizations on the Dataset

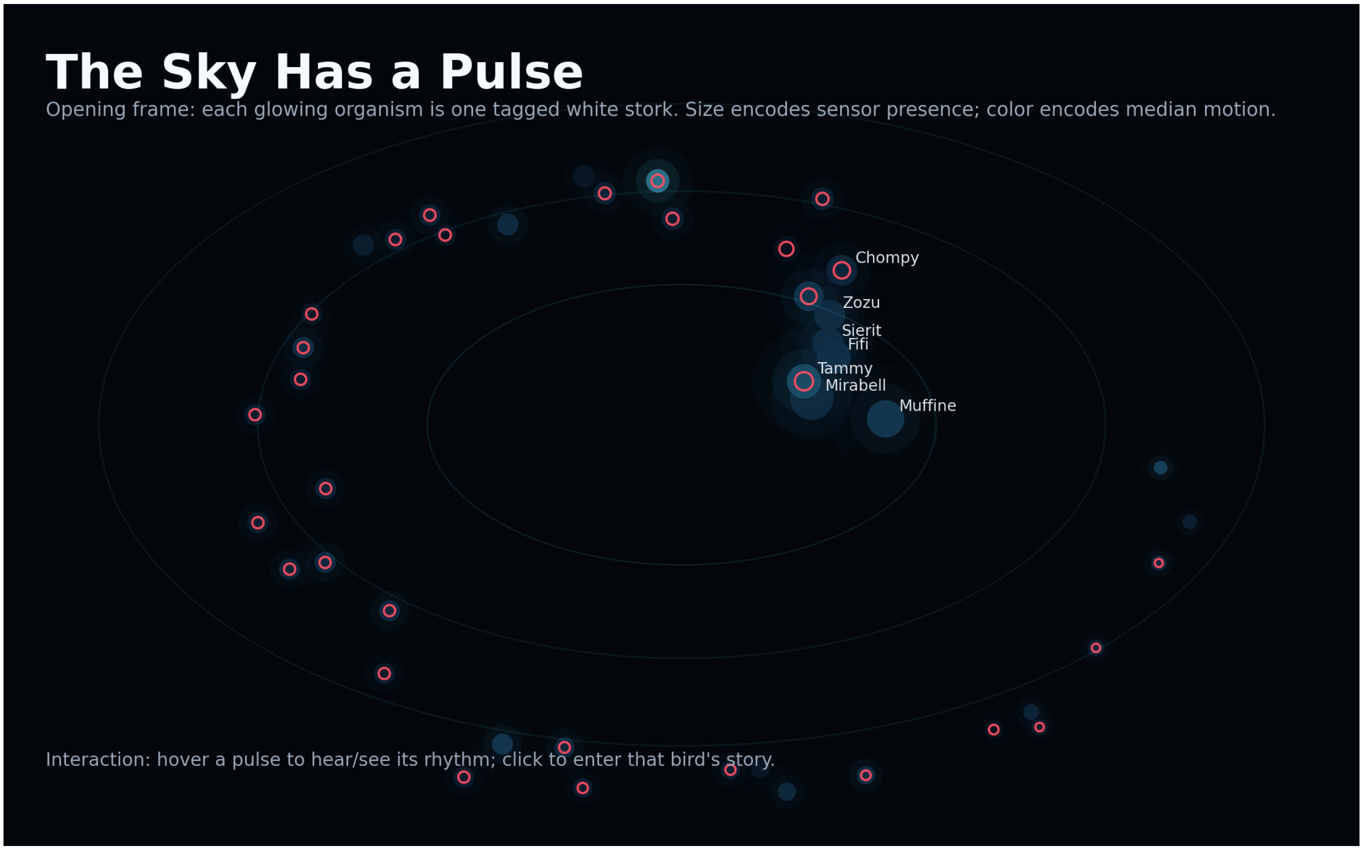


**Figure 1: Living sky overview.** Each organism-like glow represents one tagged stork. Size shows sensor presence, color shows median motion, and red outlines indicate recorded death outcomes.

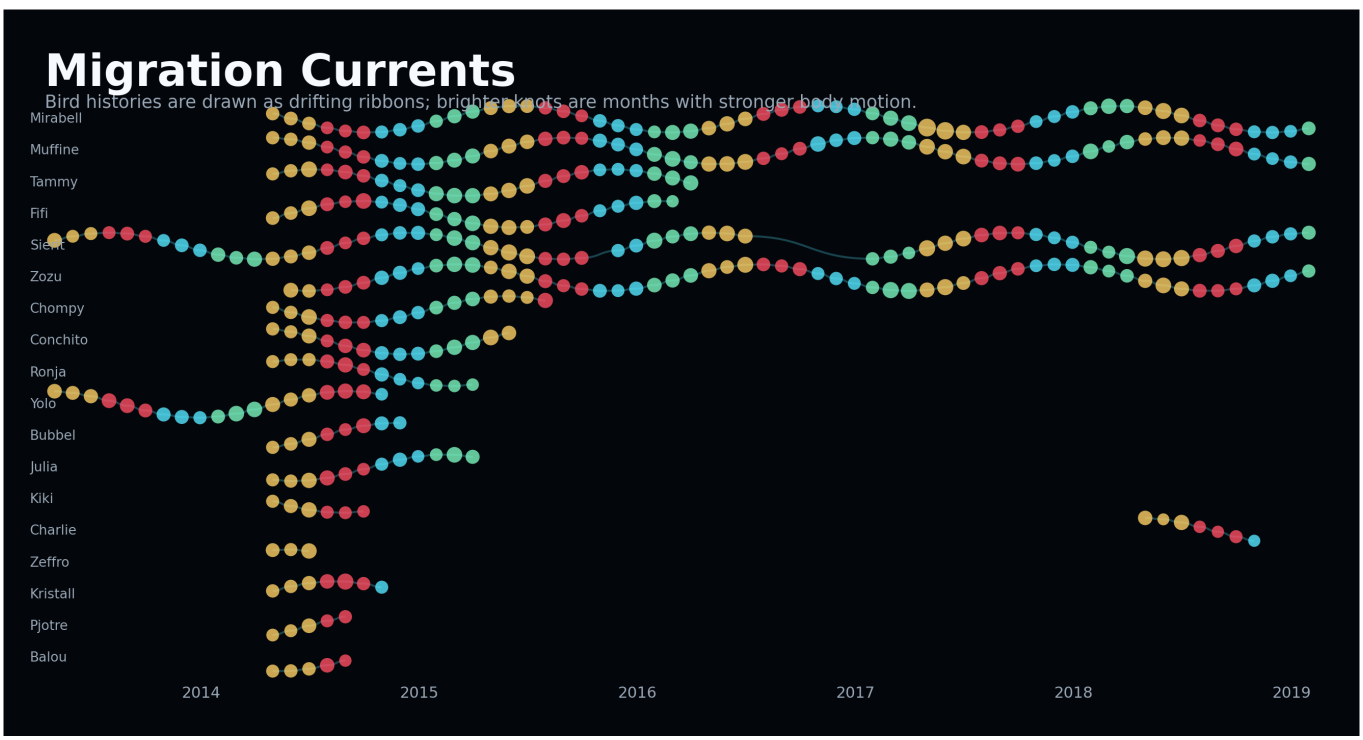


**Figure 2: Migration currents.** Individual bird histories become flowing ribbons. Bright knots mark months with stronger body motion, making seasonal activity feel like weather.


**Figure 3: Individual pulse orbit.** A selected bird's monthly activity becomes a circular rhythm, turning a time series into a bodily pulse signature.

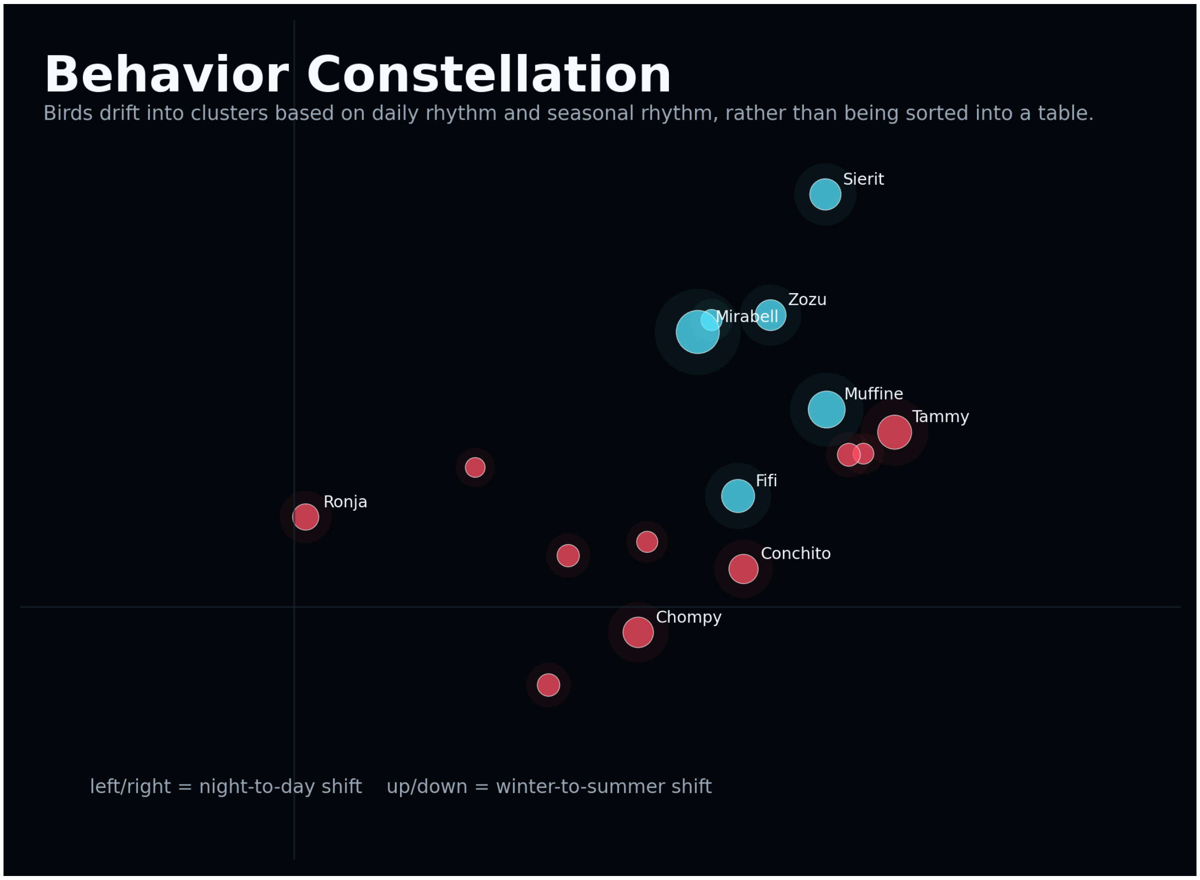


**Figure 4: Behavior constellation.** Birds are positioned by day/night and summer/winter activity differences, revealing clusters of behavioral rhythm rather than simple categories.

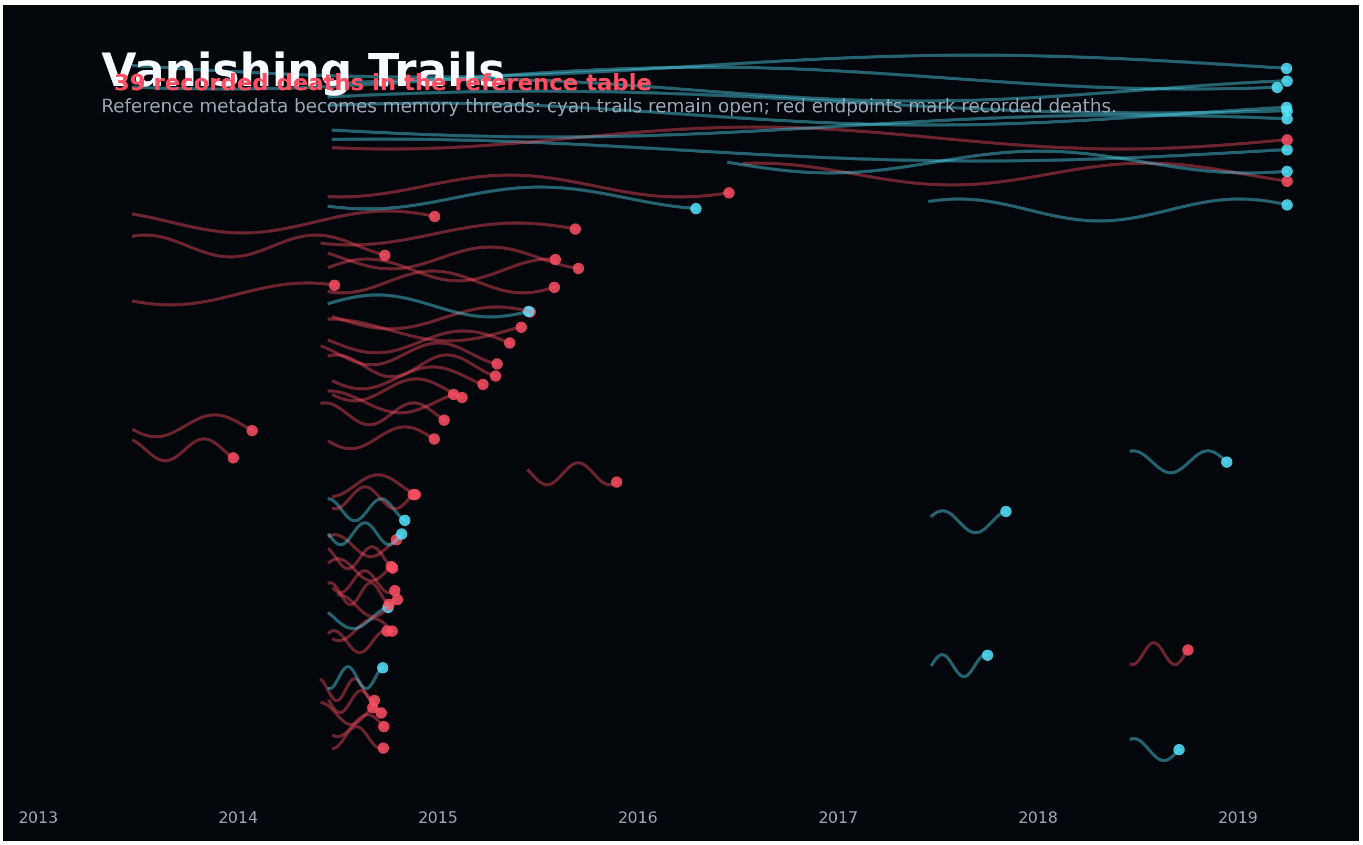


**Figure 5: Vanishing trails.** Deployment metadata becomes memory threads: cyan trails remain open, while red endpoints mark deployments ending in recorded death.

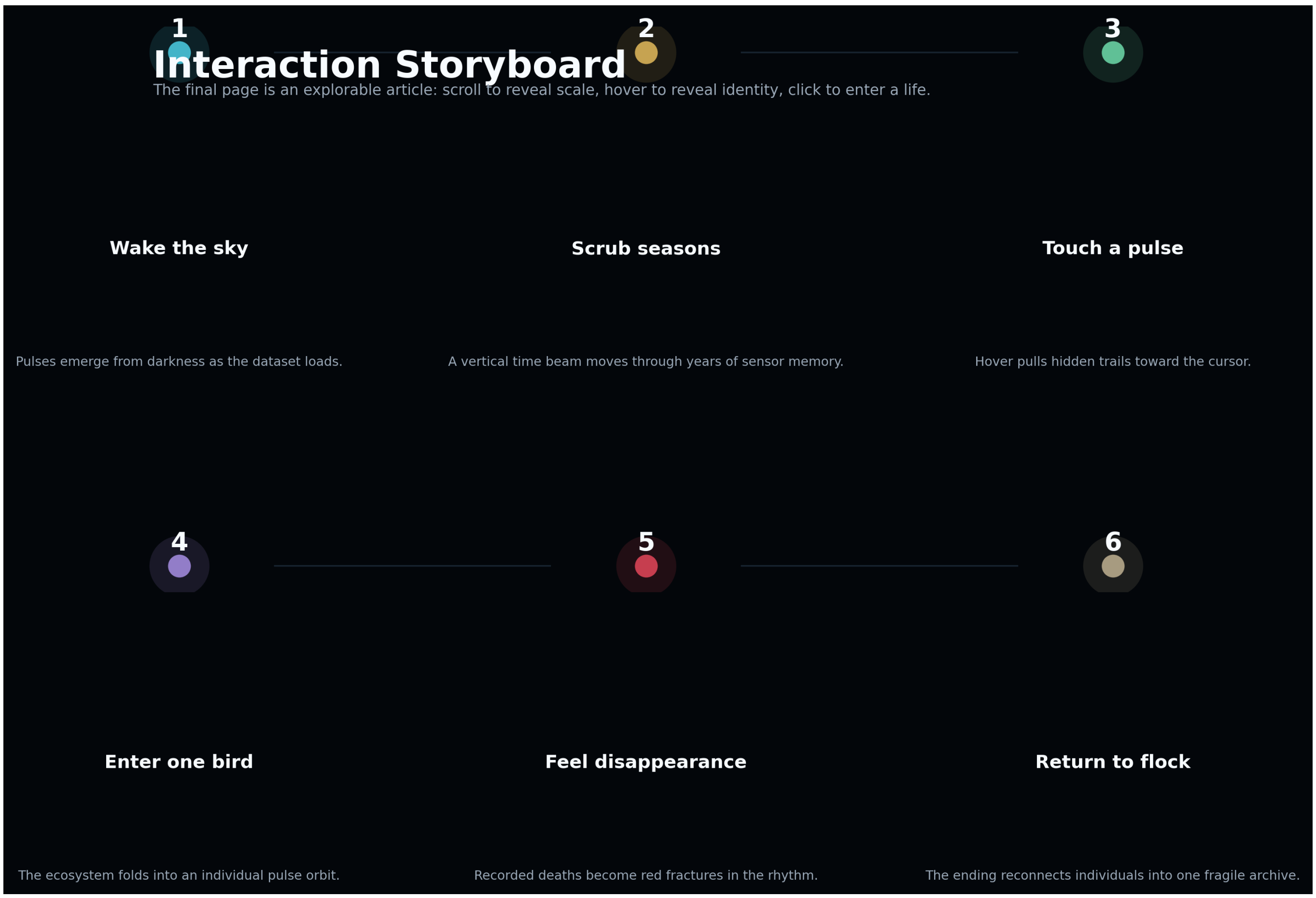


**Figure 6: Interaction storyboard.** The final explorable article will move through six scenes: wake the sky, scrub seasons, touch a pulse, enter one bird, feel disappearance, and return to the flock.